Week 6 Lab / Activity Handout

EDA Design Flow Integration | Xilinx FPGA Simulation and Synthesis

1. Purpose of this activity

Use the unchanged, verified Week 5 pipelined 3-tap FIR as a controlled Week 6 FPGA synthesis experiment. First establish functional correctness in XSim; then run two Vivado synthesis configurations, read coverage and resource/timing reports, and trace one observed timing path back to RTL. No physical FPGA board is used.

This guided activity supports attendance and participation (10%) and prepares the skills for the 45% Xilinx FPGA evaluation project.

2. Learning goals

- Reproduce both Week 5 XSim regressions, including the invalid-input bubble and signed -128 boundary.
- Keep RTL, device part, and 10 ns XDC fixed while comparing Vivado synthesis A (baseline) with B (area).
- Record measured LUT/FF/DSP48/BRAM utilization, WNS/TNS, and timing coverage without guessing outcomes.
- Identify the worst analyzed setup path and explain its mapped logic and possible RTL origin.
- Propose one evidence-backed improvement; reverify any optional RTL modification before re-synthesis.
- Recognize why synthesis timing is preliminary and Week 7 requires placement and routing.

3. Course scoring reminder

Component	Weight	Week 6 connection
Xilinx FPGA evaluation project	45%	Practices later FPGA design-flow evidence.
Cell-based digital IC evaluation project	45%	RTL correctness and report interpretation transfer to the IC flow.
Attendance and participation	10%	Guided lab, checkpoint answers, and reasoned explanations.

4. Activity schedule (180 minutes)

Time	Activity	Expected outcome
0-80 min	Slides: synthesis controls, timing coverage, resource reports and path reasoning	Understand the experimental questions.
80-90 min	Break	
90-155 min	Guided Parts A-F: XSim, modes A/B, report comparison and path diagnosis	Recorded measured evidence; C/D/E only if time allows.
155-165 min	Break	
165-180 min	Checkpoint, discussion, submission and Week 7 handoff	One report-supported optimization proposal.

5. Software-only scope and environment

- Target: Xilinx PYNQ-Z2 programmable logic, part xc7z020clg400-1; synthesis top: fir3_pipe2 (NOT a testbench).
- Clock analysis target: 10.000 ns on clk (100 MHz constraint), NOT a claim of measured or physical board-clock frequency.
- No bitstream generation, download, board programming, hardware measurement, or ASIC tapeout in this activity.
- Use a supported machine or remote environment where xvlog, xelab, xsim, and vivado are on PATH. Native macOS does not run Vivado directly.
- Extract Week6_Inclass_Lab_Week5_Aligned.zip and run commands from its extracted Week6_Inclass_Lab_Week5_Aligned/ root.

6. Files in the aligned lab package

Path	Role
rtl/fir3_pipe2.v	The canonical Week 5 pipelined FIR; simulation DUT / synthesis top fir3_pipe2.
tb/tb_fir3_pipe2.v	Self-checking main XSim testbench (bubble + final drain).
tb/tb_fir3_pipe2_signed.v	Separate signed-boundary XSim testbench.
constraints/fir3_pipe2.xdc	10 ns create_clock and reset-origin false-path constraint.
scripts/run_xsim_regression.sh	Runs BOTH XSim testbenches, checks results, and copies VCDs.
scripts/run_synth_modes.sh	Starts isolated Vivado batch processes for requested modes A-E.
scripts/synth_modes.tcl	Exact common synthesis inputs, options, and report commands.
reference/fir3_expected_outputs.txt	Canonical Week 5 test sequence and bubble/latency notes.
reference/fir3_pipe2_expected_outputs_week6.txt	Separate illustrative input sequence [1,2,3,0,0]; NOT the main testbench stimulus.

A. Classify before running tools

Mark Yes or No for whether each item participates in the simulation or synthesis workflow. A testbench is simulation-only; scripts direct tools but are not synthesizable RTL.

Item	Sim?	Synth?	Reason
rtl/fir3_pipe2.v	____	____	_____
tb/tb_fir3_pipe2.v / ..._signed.v	____	____	_____
constraints/fir3_pipe2.xdc	____	____	_____
scripts/run_xsim_regression.sh	____	____	_____
scripts/run_synth_modes.sh	____	____	_____
scripts/synth_modes.tcl	____	____	_____

B. Run both XSim regressions (Slides 3-4, 29)

Execute in the extracted package root:

```
chmod +x scripts/*.sh
./scripts/run_xsim_regression.sh
```

Check the test-specific XSim logs in logs/ and the two generated VCD files in waves/. The runner must print:

```
[RESULT] FIR3_PIPE2 TEST PASSED
[RESULT] FIR3_PIPE2 SIGNED TEST PASSED
[RESULT] WEEK6 INCLASS FIR REGRESSIONS PASSED
```

Regression	Log / waveform	Evidence to record
Main	logs/fir3_pipe2_xsim.log; waves/fir3_pipe2.vcd	PASS; reset, valid bubble, output drain.
Signed boundary	logs/fir3_pipe2_signed_xsim.log; waves/fir3_pipe2_signed.vcd	PASS; signed -128 repeated three times; final -512.

Main regression reference, after each rising edge (output holds when invalid):

Edge	valid_in	x_in	valid_out	y_out	What it demonstrates
E1	1	1	0	0	Product stage fills
E2	1	2	1	1	First accepted output
E3	0	99	1	4	Bubble: prior accepted output drains
E4	1	3	0	4	Delayed bubble; output holds
E5	1	4	1	8	Resume accepted outputs
E6	1	5	1	12	Continuous throughput
E7	0	0	1	16	Final result drains
E8	0	0	0	16	Idle output holds

In E3 the history/product FFs hold because valid_in=0, but y_out may still update from the previous valid Stage-1 values. The mathematical accepted-sample output sequence is 1, 4, 8, 12, 16. These are checked by the main testbench, not the alternative reference file.

C. Inspect waveform evidence

Display clk, rst_n, valid_in, x_in, valid_s1, valid_out, y_out, and if useful d0/d1 and p0_s1/p1_s1/p2_s1. Mark one reset check, the E3/E4 invalid bubble, and one final drain edge. For the signed test, confirm output values -128, -384, -512 (the last after the drain edge).

Observation: main PASS _____ signed PASS _____ VCD files exist _____

D. Controlled Vivado synthesis: A versus B (Slides 5-13, 29)

Do not change RTL or constraints during the A-E sweep. Run the primary A/B experiment:

```
./scripts/run_synth_modes.sh A B
```

Optional only if time permits:

```
./scripts/run_synth_modes.sh C D E
```

Mode	Synthesis option relative to baseline A	Interpretation (no guaranteed change)
A	directive default; flatten_hierarchy rebuilt; resource_sharing auto; global_retiming off	Baseline
B	-directive AreaOptimized_high	May favor area; timing may improve or worsen.
C	-flatten_hierarchy full	Limited effect possible: no instantiated child RTL modules.
D	-resource_sharing off	Few mutually exclusive operators in this constant FIR.
E	-global_retiming on	Async-reset FFs may restrict eligible retiming.

The runner starts a separate Vivado batch process in build/mode_<mode>/ for each mode and checks the completion marker and all five nonempty output files. Expect terminal [PASS] A..., [PASS] B..., then [RESULT] REQUESTED SYNTHESIS MODES COMPLETED; these mean artifacts exist, not that WNS is nonnegative.

7. Exact output files (for each mode)

Location	Purpose
logs/synth_<mode>.log	Vivado transcript, settings, warnings, errors and completion marker.
reports/<mode>/check_timing.rpt	Missing-clock / unconstrained-path / timing coverage checks.
reports/<mode>/utilization.rpt	Mapped LUT, FF, DSP48E1, BRAM utilization counts.
reports/<mode>/timing_summary.rpt	Estimated WNS/TNS and WHS/THS; includes unconstrained information.
reports/<mode>/worst10_setup_paths.rpt	Up to 10 max-delay paths; inspect reported start/endpoints and cells.
reports/<mode>/post_synth.dcp	Synthesis checkpoint; not a routed implementation.

8. Warnings and coverage are part of evidence (Slides 16-17)

Review synthesis log errors/critical warnings and unexpected optimization or inferred latches. Read check_timing.rpt BEFORE interpreting timing_summary.rpt. The supplied XDC has a 10 ns clock, no external data I/O delays, and a false-path exception from rst_n. Therefore positive WNS is not complete interface timing closure or proof of safe reset deassertion.

E. Read utilization, setup/hold and diagnose one real path

Copy actual values for A and B (Slides 14, 17-23, 29-30). Use N/A, not guessed numbers, when a value is absent; compare like-for-like report fields.

Measurement	A (baseline)	B (area)	B - A
LUT (count)	_____	_____	_____
FF (count)	_____	_____	_____
DSP48E1 (count)	_____	_____	_____
BRAM (count)	_____	_____	_____
WNS (ns)	_____	_____	_____
TNS (ns)	_____	_____	_____
WHS (ns) / THS (ns)	_____	_____	_____

What do check_timing.rpt and the unconstrained-path section say about clock existence and external I/O timing?

_____

Interpretation: WNS is the minimum setup slack of analyzed paths; TNS accumulates negative worst setup slack over failing endpoints. WHS/THS concern hold checks. Post-synthesis interconnect delays are estimates, not post-route measurements.

9. Exercise 2: actual worst-path trace (Slide 30)

Read reports/A/worst10_setup_paths.rpt (and B if helpful). Even if WNS is positive, characterize the worst analyzed path; do not invent a violation. Confirm any speculation about the two serial adders against the reported cells.

Question	Answer / actual report excerpt
Mode; clock / timing group	_____
Startpoint and endpoint	_____
Path slack; arrival / required times	_____
Mapped LUT / CARRY4 / FF cells and estimated nets	_____
Likely source RTL registers and expressions	_____
Dominant delay signature and supporting evidence	_____

Candidate logic cone in the original source: p0_s1/p1_s1/p2_s1 -> s0_s1 = p0_s1 + p1_s1 -> y_s1 = s0_s1 + p2_s1 -> y_out FF. It is a hypothesis until the actual timing report confirms it. Unused d2 registers may be optimized away.

F. Engineering recommendation and optional re-verification (Slides 24-31)

REQUIRED: Propose one change based on the path you actually observed. State how it might affect LUT/FF counts, WNS, valid alignment, or latency. "Area mode is better" is not justified unless measurements support the specific metric.

Decision worksheet	Student response
Observed path / report evidence	_____
One proposed RTL or constraint-correction action	_____
Predicted resource or timing trade-off	_____
How you will verify correctness and timing	_____

OPTIONAL EXTENSION: If instructed to implement and measure an RTL change, preserve the original baseline and work on an isolated candidate copy. For the slide example, register s0_s1 and a matching delayed p2_s1, add a corresponding valid stage, and accept one extra cycle of output latency. Update both testbenches expected cycle timing (including reset/bubbles/drain and signed -128) before requiring both regressions to PASS; then re-synthesize at the SAME 10 ns constraint. The distributed scripts do not supply a ready-made modified RTL or modified testbench.

Numerical safety: signed input is 8-bit (-128 ... 127); coefficients [1,2,1] give -512 ... +508, requiring signed 10-bit outputs. Never narrow or change signed arithmetic without proving its range and rerunning boundary tests.

10. Evidence checklist: same requirements as Slide 32

Submission item	Required?	Evidence or path recorded
Week 5 RTL and both testbench filenames (identify exact code version)	Yes	_____
Both XSim PASS markers, and selected waveform/VCD evidence	Yes	_____
A/B LUT, FF, DSP48, BRAM measurements from actual reports	Yes	_____
A/B WNS, TNS and timing coverage / unconstrained-I/O note	Yes	_____
One ACTUAL worst path: startpoint, endpoint, mapped cells, RTL trace	Yes	_____
One report-supported optimization proposal and expected trade-off	Yes	_____
Implemented RTL change + rerun evidence	Optional	_____
Extra C/D/E results	Optional	_____

11. Exit discussion and Week 7 handoff

Why might B use fewer LUTs but have worse WNS? _____

Can positive WNS alone demonstrate complete timing closure? _____

Why rerun both regressions after an RTL change? _____

Week 7 will use the synthesized checkpoint for optimization, placement, routing and post-route STA. Compare estimated synthesis WNS with routed WNS under the same 10 ns constraint; Week 6 has no board execution.